

WiP: UnderTone: Frequency Lock Encoding on COTS Wi-Fi Hardware

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Abstract

Modern wireless systems reserve excess synchronization bandwidth to ensure reliability under worst-case conditions. Recent foundational work introduced Frequency Lock Encoding (FLE), a protocol-agnostic signaling layer that opportunistically exploits this margin by embedding data through small, controlled carrier frequency offsets (CFO). While FLE has been successfully validated using software-defined radios (SDRs) on single-carrier waveforms, deploying it on commercial off-the-shelf (COTS) Wi-Fi devices has historically been hindered by locked physical layers and highly unpredictable Medium Access Control (MAC) timing. This work-in-progress presents UnderTone, a fully software-defined FLE subprotocol implemented on standard ESP32-C5 microcontrollers. By reverse-engineering undocumented fractional- N divider registers to induce precise CFO shifts, and modeling the bursty Wi-Fi medium as an erasure channel protected by Reed-Solomon coding, UnderTone completely bypasses the need for custom SDR hardware and brings this resilient physical-layer subprotocol to consumer IoT devices.

CCS Concepts

• **Networks** → **Network protocol design**; *Wireless local area networks*; • **Computer systems organization** → Embedded systems.

Keywords

Wireless Communications, Subprotocol, Frequency Lock Encoding, IoT

ACM Reference Format:

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1 Introduction

Wireless links face unpredictable analog disturbances, so protocol designers provision synchronization mechanisms that reserve extra bandwidth for worst-case impairments—conditions rarely encountered in normal operation. Recent work introduced Frequency Lock Encoding (FLE), a signaling technique that treats this unused synchronization capacity as a secondary resource [5]. FLE induces small, controlled frequency offsets into a primary transmission to embed a parallel data stream, while remaining within the synchronization correction limit of the underlying protocol.

The mathematical foundation and practical feasibility of FLE were initially established for single-carrier waveforms [5]. Through simulation and over-the-air SDR experiments, FLE achieved secondary data rates exceeding 900 kbps with negligible degradation to the primary link. Prior work has also explored similar CFO overlays for covert channels [2]. While these systems established feasibility and evaluated key parameter tradeoffs, they were realized on SDR platforms with substantial physical-layer access.

Transitioning this capability from SDR environments to commercial off-the-shelf (COTS) wireless systems presents substantial architectural challenges. Modern consumer Wi-Fi chipsets do not expose raw IQ samples or internal PLL signals to the application layer, making standard FLE modulation and detection seemingly impossible without hardware modifications. Furthermore, the IEEE 802.11 MAC layer distorts Wi-Fi frame transmission timing, which is relied upon for FLE.

In this work, we present UnderTone (see Figure 1), a resilient FLE system built entirely in software on the COTS ESP32-C5 microcontroller. We overcome physical-layer restrictions by mapping undocumented hardware registers to CFO generation and extraction. Furthermore, we address MAC layer unpredictability by modeling the bursty wireless medium as an erasure channel, protected by Reed-Solomon coding and block interleaving.



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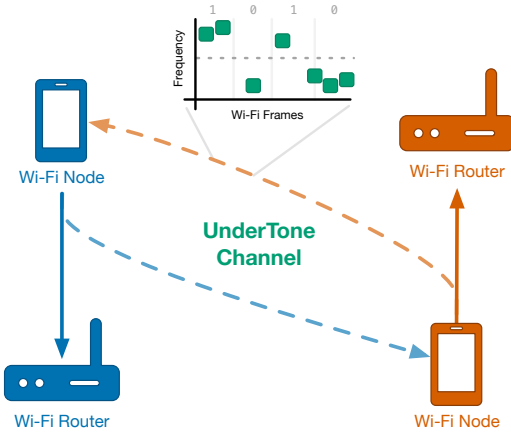


Figure 1: UnderTone allows devices to simultaneously communicate with the primary receiver and another UnderTone node, even across networks

2 COTS Hardware Adaptation

To transition FLE from SDRs to COTS devices, we implement UnderTone on the ESP32-C5 microcontroller using exclusively software-based modifications. Modern Wi-Fi receivers correct oscillator mismatches via an automatic frequency correction (AFC) loop [4] that cross-correlates training symbols at the start of each 802.11 burst. We leverage this process to enable the subprotocol channel.

By default, the ESP32 family does not expose CFO statistics to the application layer. However, we found that during the AFC process, the ESP32 hardware records the CFO of each incoming packet into a dedicated, undocumented hardware register. By accessing this register at the correct stage of the reception pipeline, the UnderTone receiver can accurately extract the FLE signal.

For the transmitter, the ESP32 silicon contains a fractional- N divider responsible for tuning the local oscillators [3]. Using an SDR, we measured the CFO induced by various divider control register settings and experimentally mapped the register-to-CFO relationship. These induced offsets are highly reliable and repeatable, allowing the transmitter to precisely tune the oscillator across a range of frequency values to transmit symbols.

While the described hardware adaptation bypasses the need for custom SDRs, it introduces two significant challenges. First, FLE requires a steady stream of primary link data upon which to modulate the subprotocol. Wi-Fi traffic, however, is frame-based and inherently bursty. Second, MAC layer complexities make the exact timing of outbound traffic highly unpredictable, so even a burst-tolerant subprotocol would not know when the underlying primary link will transmit. These challenges necessitate an entirely new subprotocol design.

3 Subprotocol Design

Implementing a subprotocol over COTS Wi-Fi hardware introduces fundamental challenges. Due to the complexities of the MAC layer, including channel sensing, RTS/CTS exchanges, retry mechanisms, packet aggregation, and management frames [4], the exact composition and scheduling of outbound traffic are highly unpredictable. Furthermore, this MAC control logic is not exposed to the end user. UnderTone must therefore operate opportunistically above this primary MAC layer, using whatever carrier packets it can, yielding a channel model with unique characteristics.

3.1 MAC Contention and Rate Adaptation

UnderTone navigates the MAC timing constraint via the ESP32 SDK’s `tx_done` callback, which asserts when the lower MAC finishes processing a transmit descriptor—an event that typically, but not necessarily, corresponds to completed RF emission. Under link saturation, the MAC may silently discard frames while still reporting success. The flag, therefore, serves as a pacing signal and an upper bound for physical layer transmissions.

Rather than enforcing a rigid, one-to-one mapping between an FLE symbol and a discrete Wi-Fi frame, UnderTone derives a virtual symbol clock from this pacing rate. The optimal FLE symbol period, T_{sym} , is dynamically calculated from a moving average of `tx_done` arrivals, treating the callback stream as a throughput estimate with known upper-bound bias. This time-averaged rate adaptation ensures that the receiver maintains a predictable observation window, remaining resilient to the inherently bursty nature of the underlying traffic without requiring per-frame transmission confirmation. To characterize burstiness, UnderTone additionally tracks the coefficient of variation (CV) of inter-arrival times, enabling adaptation to diverse Wi-Fi traffic patterns.

3.2 The Dual-Threat Channel

By decoupling the symbol clock from primary packets, UnderTone treats the wireless medium as a “dual-threat” channel. The system must simultaneously resolve: (1) traditional analog impairments (AWGN, multipath fading, and oscillator drift) causing bit errors; and (2) MAC-layer timing erasures, where the absence of primary transmissions due to contention or idle windows leaves no carrier packets to modulate the UnderTone signal. To overcome this, UnderTone employs Reed-Solomon (RS) forward error correction (FEC) using Berlekamp-Massey and Forney decoding algorithms [1]. This dynamically shares the row parity budget (P) across $2E + e \leq P$, consuming one parity byte per erasure (e) and two per random RF error (E).

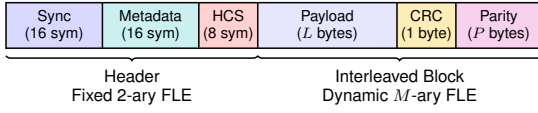


Figure 2: UnderTone packet structure

3.3 Packet Architecture

To facilitate reliable recovery over this unique channel, UnderTone utilizes a specifically engineered packet architecture, illustrated in Figure 2. Each packet begins with a fixed 40-symbol header transmitted using robust binary FLE. A 16-symbol synchronization word recovers the transmitter’s virtual clock and symbol levels, followed by metadata containing a 4-symbol Modulation and Coding Scheme (MCS) field, a 4-symbol parity profile field, and an 8-symbol payload length field, protected by an 8-symbol Header Check Sequence (HCS). Following the header, the hardware shifts into the MCS-specified M -ary constellation. The payload, a CRC checksum, and the Reed-Solomon parity block are combined into a 2D matrix and block-interleaved prior to transmission. This interleaving distributes contiguous multi-symbol erasures from absent carrier traffic across RS codewords, ensuring the payload remains recoverable.

Because the primary Wi-Fi traffic acts merely as a carrier signal, UnderTone requires only a general approximation of its pacing rate and is agnostic to packet contents. This decoupling means the subprotocol data need not be destined for the primary frame’s recipient: an UnderTone node can transmit to any associated node, a node on a neighboring network, or even an unassociated passive listener.

4 Preliminary Evaluation

To validate UnderTone, we first established an over-the-air baseline. Under continuous Wi-Fi medium saturation, the subprotocol achieved 100% packet reception reliability in a laboratory setting with no primary protocol degradation.

Because real-world Wi-Fi traffic is highly irregular, we develop a high-fidelity simulator using different traffic patterns to evaluate rate adaptation under constant, Poisson, and bursty profiles. By simulating a `tx_done` callback stream, the transmitter estimates both the mean packet rate and inter-arrival-time variability. These statistics drive a variance-aware MCS selection algorithm, selecting the optimal modulation scheme in 94.1% of simulated scenarios.

We deploy the algorithm on ESP32-C5 hardware and evaluate the complete UnderTone system over-the-air in a laboratory environment across all three traffic patterns. In every scenario, the transmitter autonomously selected the optimal MCS, driving bit errors low even without FEC. The results and effect of the FEC packet structure are summarized in

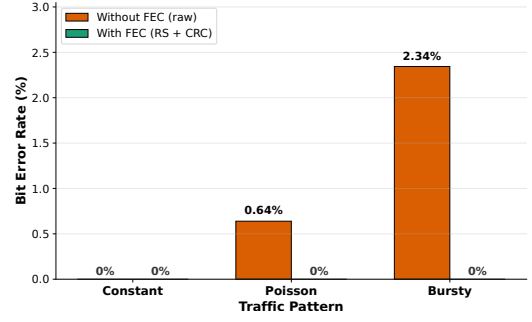


Figure 3: BER of UnderTone over-the-air transmissions across Wi-Fi traffic patterns

Figure 3; as expected, Poisson and bursty patterns introduced more erasures, but all were absorbed by RS error correction.

5 Future Work

Moving forward, our immediate focus is to fully characterize the operational boundaries of this system. This includes evaluating UnderTone under real applications and their traffic patterns, examining its performance in highly congested environments with significant MAC-layer contention, and developing a device discovery mechanism for UnderTone-enabled nodes. Once these capabilities are thoroughly understood, we will build a complete application leveraging this technology. Potential applications include watermarking wireless signals to enhance security, enabling cross-network spectrum coordination through the side channel, and offloading network coordination and control information from the primary data plane to improve overall channel utilization and network efficiency.

Acknowledgments

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References

- [1] Elwyn R. Berlekamp. 1968. *Algebraic Coding Theory*. McGraw-Hill, New York.
- [2] Jiska Classen, Matthias Schulz, and Matthias Hollick. 2015. Practical Covert Channels for WiFi Systems. In *Proceedings of the 2015 IEEE Conference on Communications and Network Security (CNS)*. IEEE, Florence, Italy, 209–217. doi:10.1109/CNS.2015.7346848
- [3] Espressif Systems. 2025. *ESP32-C5 Technical Reference Manual* (version 1.1 ed.). https://documentation.espressif.com/esp32-c5_technical_reference_manual_en.pdf
- [4] IEEE. 2021. *IEEE Standard for Information Technology Telecommunications*. Technical Report IEEE Std 802.11-2020 (Revision of IEEE Std 802.11-2016). Institute of Electrical and Electronics Engineers. doi:10.1109/IEEESTD.2021.9363693
- [5] Joshua Montierth, Michael Rice, and Philip Lundrigan. 2026. Frequency Lock Encoding. arXiv:2606.20813 [cs.NI] <https://arxiv.org/abs/2606.20813> Accepted to IEEE MILCOM 2026.